RDS

What is AWS RDS?

Amazon RDS is a **managed database service** provided by **Amazon Web Services**. It allows you to **set up, operate, and scale relational databases in the cloud** without worrying about infrastructure management.

Why use RDS?

Instead of manually installing databases on EC2, RDS handles:

- Server provisioning
- OS patching
- Database setup
- Backups & recovery
- Scaling
- High availability

So you focus on your **application**, not database maintenance.

Supported Database Engines

RDS supports multiple relational database engines:

- MySQL
- PostgreSQL
- MariaDB
- Oracle
- Microsoft SQL Server
- **Amazon Aurora** (AWS-native DB, faster & cheaper)

Key Features

1. Automated Backups

- Automatically backs up your database
- Point-in-time recovery (restore to any second)

2. Multi-AZ Deployment (High Availability)

- Data is replicated to a standby instance in another Availability Zone
- Automatic failover if primary DB fails

3. Read Replicas (Scaling Reads)

- Create copies of your DB for read-heavy applications
- Improves performance

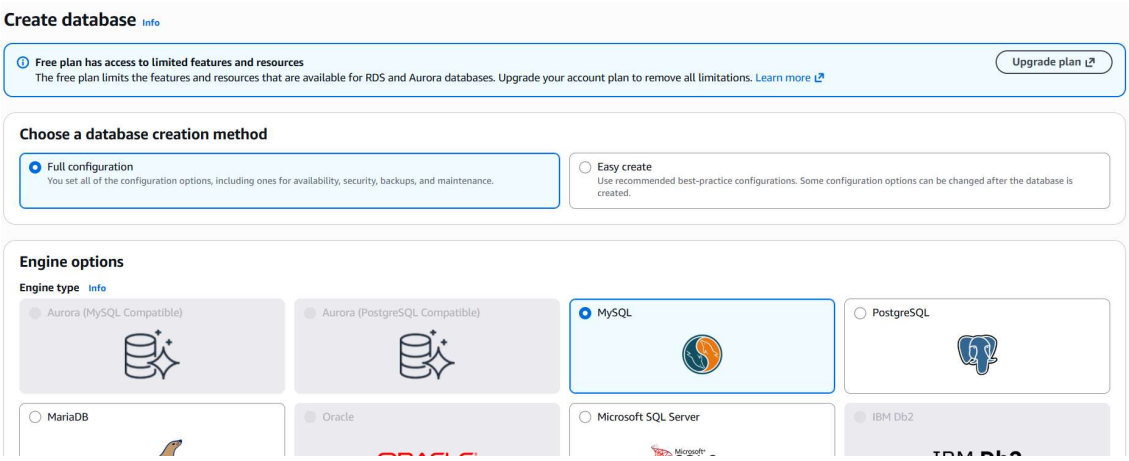
4. Security

- VPC isolation
- IAM authentication
- Encryption (at rest & in transit using KMS)

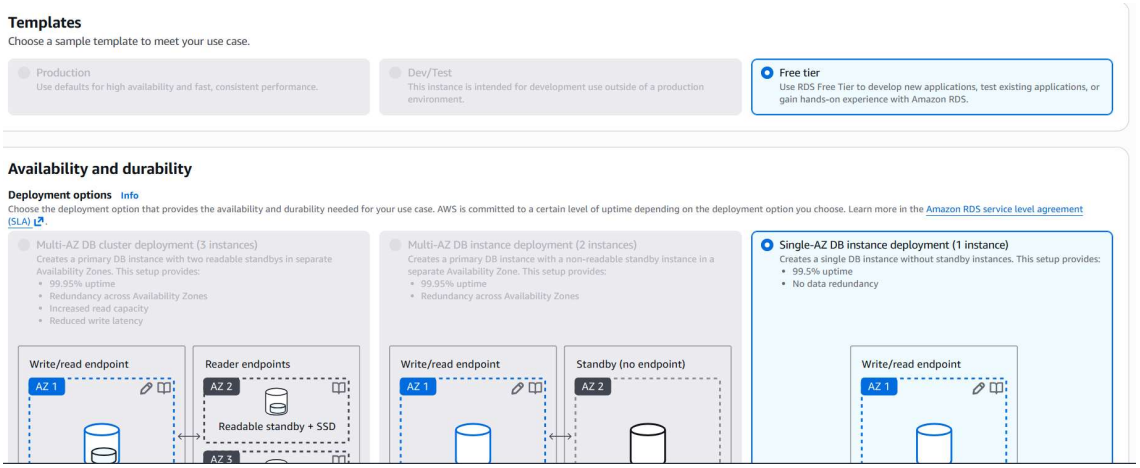
5. Scaling

- Vertical scaling (increase instance size)
- Storage auto-scaling

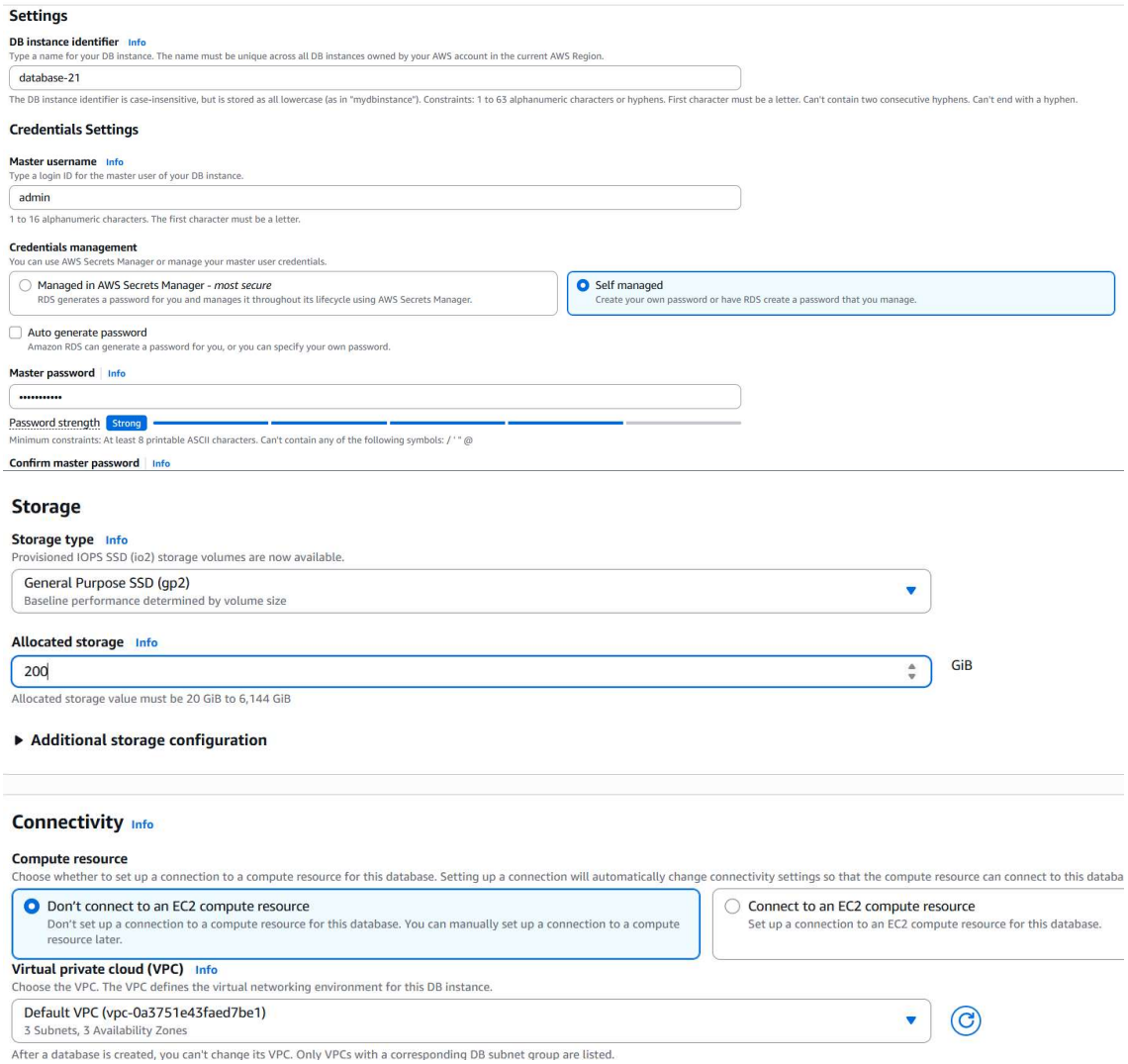
First to create database by using "**MySQL**"



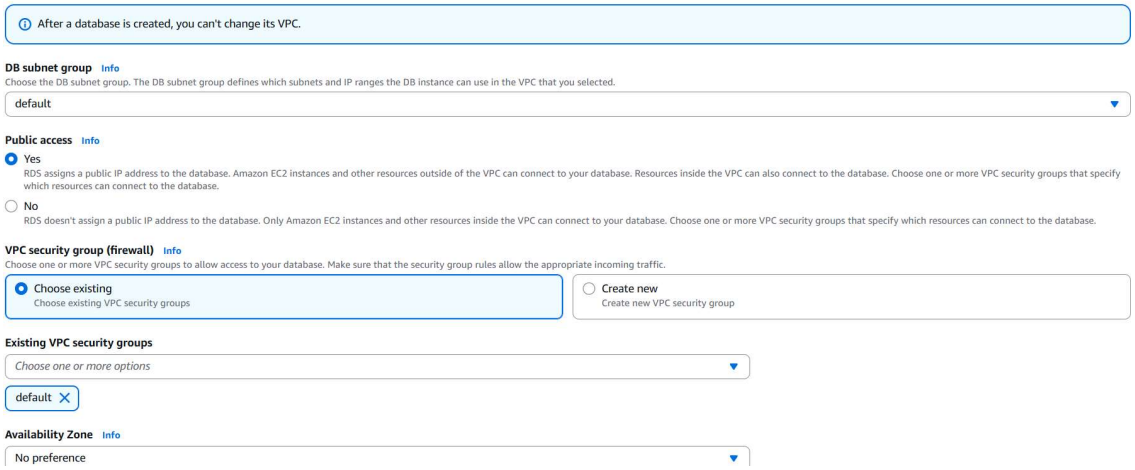
Next to add the Template for our database



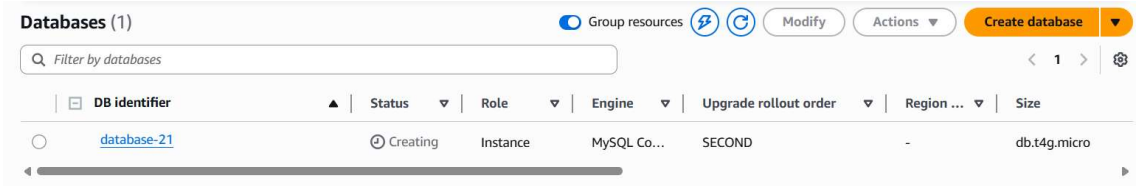
Add the name,username and password for our database and allocate storage capacity



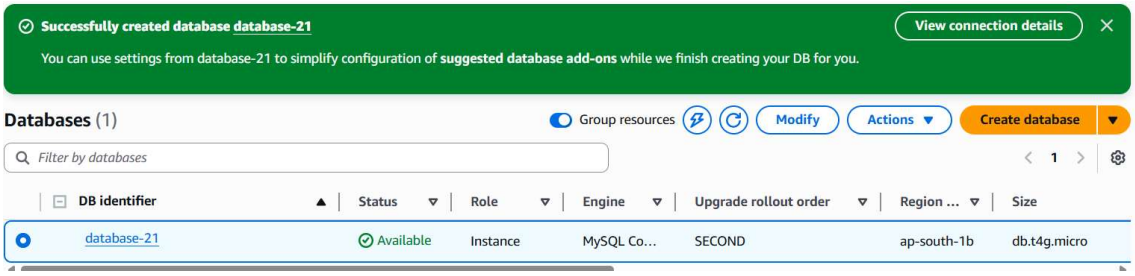
Allow public access



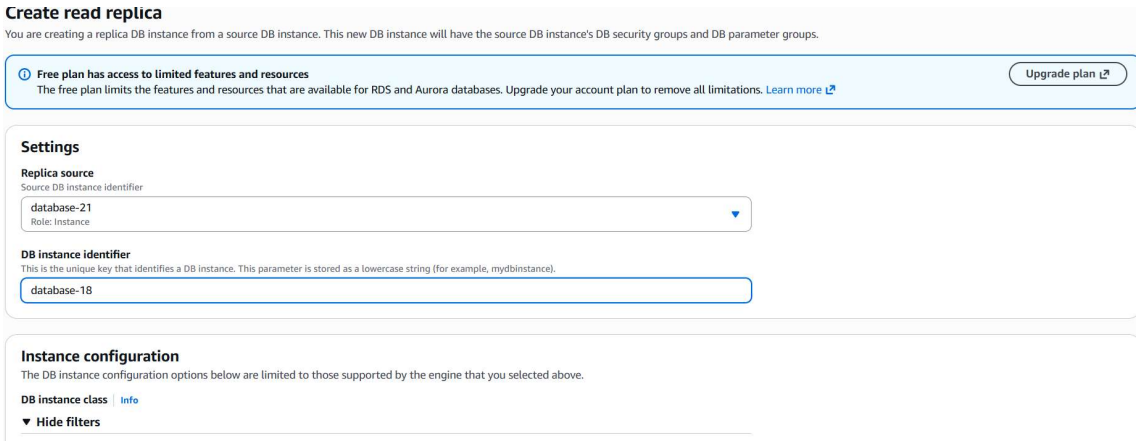
My database is creating



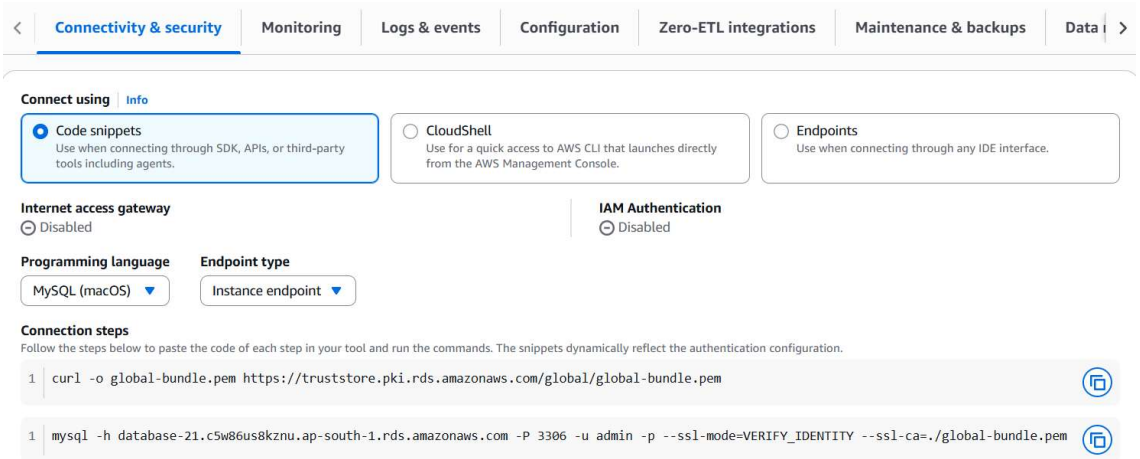
It is available now



Create read replica for our database



- This is our end point of database



- After creating database,we have to create instance by using ubundu machine

```
23 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-46-255:/home/ubuntu# apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfc
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libme
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common m
```

- To connect the DATABASE by using the endpoint(that having username and password)

```
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-46-255:/home/ubuntu# mysql -h database-21.c5w86us8kznu.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 8.4.7 Source distribution

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```

- SHOW DATABASES;→for list the databases

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.01 sec)

mysql>
```

- CREATE DATABASE hari;→ for create the database by the name of hari

```
mysql> CREATE DATABASE hari;
Query OK, 1 row affected (0.02 sec)

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| hari               |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)
```

CREATE TABLE Customer(

CustomerID INT PRIMARY KEY,

FirstName VARCHAR(50),

LastName VARCHAR(50),

Country VARCHAR(50),

Age INT CHECK (Age >= 0 AND Age <= 99),

Phone int(10)

);

- For create table with those columns

DESC customer;→for list the table

```
mysql> CREATE TABLE Customer(     CustomerID INT PRIMARY KEY,     FirstName VARCHAR(50),     LastName VARCHAR(50),     Country VARCHAR(50),     Age INT CHECK (Age >= 0
 AND Age <= 99),     Phone int(10) );
ERROR 1050 (42S01): Table 'Customer' already exists
mysql> DESC Customer;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CustomerID | int         | NO   | PRI | NULL    |       |
| FirstName  | varchar(50) | YES  |     | NULL    |       |
| LastName   | varchar(50) | YES  |     | NULL    |       |
| Country    | varchar(50) | YES  |     | NULL    |       |
| Age        | int         | YES  |     | NULL    |       |
| Phone      | int         | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
```